Stability without concealment

When does grouping correlated inputs improve explanation stability without hiding useful information?

THE PROBLEM

Similar predictions can produce different feature rankings after retraining.

THE TRAP

Shorter lists and random grouping can also appear more stable.

OUR TEST

Compare matched groups; measure stability plus fidelity and cancellation.

Credit benchmark • interpretable baseline • controlled comparisons • eight weeks

[Student name] | [Roll number] | B.Tech IT, DTU | Prof. Rahul Gupta

What the literature establishes — and what we will test

SHAP + dependence	Credit depends on model, background and explanation game. Lundberg & Lee; Aas et al.
Disagreement + aggregation	Agreement is a research problem; fusion already exists. Krishna et al.; Bhatt et al.
Grouped explanations	groupShapley and recent robustness work are close prior art. Jullum et al.; Hendro et al. (2025)

Proposed gap: test stability beyond matched random grouping, together with information loss and known-ground-truth fidelity.

Sources: SHAP [S01]; disagreement [S03]; groupShapley [S04]; ITNAC 2025 [N03]; agreement 2026 [N04]

Methodology and controlled evaluation

01

Split + preprocess

Fit only on training data

02

Retrain + explain

Same held-out people

03

Group + controls

Match group sizes exactly

04

Evaluate

Stability AND retained detail

MEASURE

RBO stability • matched-random difference
Cancellation • synthetic attribution error
AP / ROC-AUC / Brier • runtime

CONTROL

No grouping • same-size random groups
Known additive + interaction cases
Fixed model/output/background per comparison

Minimum scope: synthetic data + UCI credit; logistic + one tree family. HELOC is conditional on verified access.

Pilot results and eight-week validation plan

11 groups; matched group sizes

Controlled RBO difference: +0.061

30,000 rows • 24k train / 6k test
AP 0.489 • AUC 0.713 • Brier 0.148
9 checks pass; deterministic replay matches

EIGHT-WEEK VALIDATION

W1–2 Data + synthetic oracle
W3–4 Tree SHAP + matched controls
W5–6 Fidelity + replication
W7–8 Paper + advisor review

Limit: one split, logistic model only. No significance, causal recourse or publication claim.